

 ज्ञानेन प्रकाशते जगत् INDUS UNIVERSITY				INDUS INSTITUTE OF TECHNOLOGY & ENGINEERING Constituent Institute of Indus University				
Subject: Software Engineering and Project Management								
Program: B. Tech Computer Science & Engineering				Subject Code: CS0601				Semester: VI
Teaching Scheme (Hours per week)				Examination Evaluation Scheme (Marks)				
Lecture	Tutorial	Practical	Credits	University Theory Examination	University Practical Examination	Continuous Internal Evaluation (CIE)- Theory	Continuous Internal Evaluation (CIE)- Practical	Total
03	00	02	04	40	40	60	60	200

Course Objectives:

1. Be successful professionals and entrepreneur in the field with solid fundamental knowledge of software engineering and Project Management.
2. Utilize and exhibit strong communication and interpersonal skills development, as well as professional and ethical principles when functioning as members and leaders of multi-disciplinary teams.
3. Apply their foundations in software engineering to adapt to readily changing environments using the appropriate theory, principles, and processes.
4. To gain the techniques and skills on how to use modern software testing tools to support software testing projects that lead them to employability in the industry.
5. To familiarize Project Management framework and Tools Contents.

CONTENTS

UNIT-I

Introduction to Software and Software Engineering

[12 hours]

Introduction to Software Engineering-Software, Evolving role of software, characteristic of software, Three “R”-Reuse, Reengineering and Retooling, SDLC, various Software Process Models, waterfall model, spiral model, incremental model, RAD model, Agile Development

UNIT-II

Software Requirement Analysis & Specification (SRS) and basic of testing

[12 hours]

SRS, quality of good SRS, Types of Requirements, Feasibility Study, Requirement Analysis, requirement engineering task, design concepts abstraction, modularity, information hiding, functional independent cohesion, coupling, testing, types of testing, black box, white box testing. Software design concepts and Design principals, Architectural Design, Component Level

Design, Function Oriented Design and Data flow Diagram, drawing rules, Leveling of DFD
Level 0, Level 1, Level 2

UNIT-III

[12 hours]

Software Project Management

Project Management - Project planning, 4 P of management, W5HH Principle, Metrics for process and project- Product metrics, Process metrics, Project metrics, LOC, FUNCTION POINT

Project Estimation techniques- Empirical estimation techniques, Heuristic techniques, COCOMO model, Project scheduling- methods, Work breakdown structure, Task network analysis (ACTIVITY NETWORK), Gantt chart, PERT, CPM

UNIT-IV

[12 hours]

Management

Risk Management – software risk, risk identification, risk projection, risk refinement, risk mitigation and monitoring,

Quality Management-quality control vs quality assurance, software review, types of review, reliability, and availability.

Change management-configuration management, change control, version control.

Course Outcomes:

At the end of this subject, students should be able to:

1. Apply use of knowledge of Software Life Cycle to successfully implement the projects in the corporate world.
2. Implement Project Management Processes to successfully complete project in IT industry.
3. Apply the concept of Functional Oriented and Object-Oriented Approach for Software Design.
4. Recognize how to ensure the quality of software product, different quality standards and software review techniques.
5. Identify the Inputs, Tools, and techniques to get the required Project deliverable and Product deliverable using Knowledge areas of Project Management and Apply various testing techniques and test plans and
6. Will be Able to understand modern Agile Development and Service Oriented Architecture Concept of Industry.

Textbooks:

1. Roger S. Pressman, Software engineering- A practitioner's Approach, McGraw-Hill International Editions.

Reference Books:

1. Managing Information Technology Project, 6edition, by Kathy Schwalbe, Cengage Learning publication.
2. Information Technology Project Management by Jack T Marchewka Wiley India publication.
3. Software Engineering 3rd edition by KK Agrawal, Yogesh Singh, New Age International publication.
4. Software Engineering Project Management by Richard H. Thayer Wiley India Publication.
5. Software Engineering for students: A Programming Approach by Douglas Bell, Pearson publication.

Web Resources:

1. <https://nptel.ac.in/downloads/106105087/>
2. <https://nptel.ac.in/courses/106101061/>

LIST OF EXPERIMENTS

Experi ment. No.	Title	Learning Outcomes
1	Study of different software engineering models. Waterfall model, Prototype model, Incremental model, Spiral model and RAD Models, Scrum Model. Select any application for which software development process can be defined and find out which process model will be more suitable for application.	CO1,CO5
2	Prepare Software Requirement Specification (SRS) document for chosen application	CO1,CO2
3	Draw E-R diagram and class diagram for chosen application	CO3

4	Write Use case Scenario and draw Use Case Diagram for chosen application.	CO3
5	Draw Activity diagram for selected application.	CO3
6	Draw sequence diagram for selected application.	CO3
7	Draw Data Flow Diagram (DFD) for selected application	CO3
8	Apply FP oriented estimation model for selected application.	CO4
9	Study of various software testing methods and design test cases for selected application	CO5
10	Study of any two Open source tools in DevOps for Infrastructure Automation, Configuration Management, Deployment Automation, Performance Management, Log Management and Monitoring	CO6